

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 12

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Score:	<u> </u>	Date:	<u> </u>
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Lab # 12: To configure route-based IPSec VPN Using Pre-shared Key Authentication

Objective: The objective of this lab is to configure route-based IPSec VPN Using Pre-shared Key Authentication **Duration: 3 Hours**

Expected Outcomes: By the end of this lab session, students will be able to:

1. To configure a route-based IPSec VPN using pre-shared key authentication between FW_A and FW_B to establish a secure communication tunnel.
2. To verify successful connectivity and data exchange between users on Network A and Network B through the established IPSec tunnel.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in [Figure 1](#), FW_A connects network A to the Internet and FW_B connects network B to the Internet. The networking requirements are as follows:

- Network A (10.1.1.0/24) connects to GE1/0/3 of FW_A.
- Network B (10.1.2.0/24) connects to GE1/0/3 of FW_B.
- FW_A and FW_B are reachable to each other.

The purpose of this networking is to set up a route-based IPSec tunnel between FW_A and FW_B and to enable users on network A and network B to communicate.

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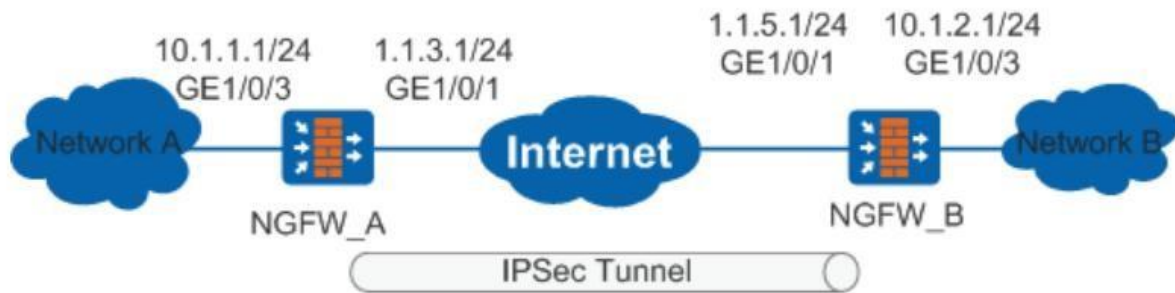


Figure 1 Networking diagram for configuring IPSEC Tunnel

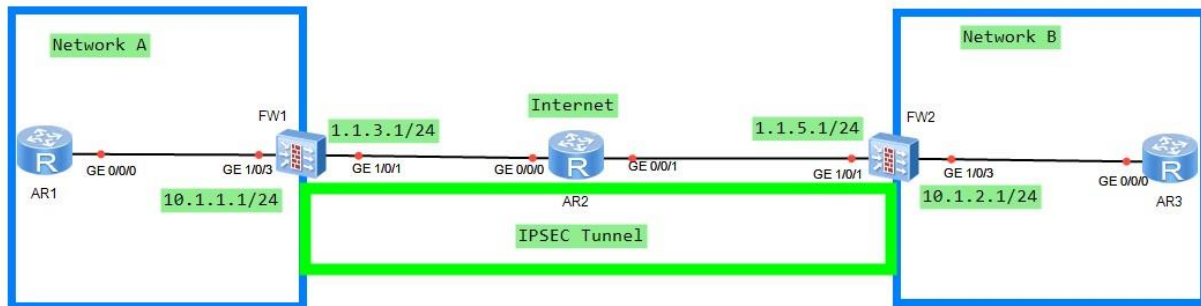


Figure 2 Networking Topology of IPSEC Tunnel

Item	Data
FW_A	Interface number: GE1/0/3 address: 10.1.1.1/24
	Interface number: GE1/0/1 address: 1.1.3.1/24
	IPSec tunnel interface configuration Source IP: 1.1.3.1 Destination IP: 1.1.5.1 IP address: any address that does not conflict with other IP addresses
	IPSec profile configuration Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP Peer ID type: any

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FW_B	Interface number: GE1/0/1 IP address: 1.1.5.1/24
	Interface number: GE1/0/3 IP address: 10.1.2.1/24
	IPSec tunnel interface configuration Source IP: 1.1.5.1 Destination IP: 1.1.3.1 IP address: any address that does not conflict with other IP addresses
	IPSec profile configuration Authentication type: pre-shared key
Item	Data
	Pre-shared key: Test!1234 Local ID: IP Peer ID: any

Configuration Roadmap

The roadmap for configuring FW_A is similar to that of FW_B:

1. Configure interfaces.
2. Configure security policies to permit packets between specified subnets.
3. Create a static route to the peer end.
4. Configure IPSec tunnel interfaces and routes to these interfaces.
5. Configure the IPSec profile, including basic IPSec profile information and proposal parameters for security association (SA) negotiation.

Procedure

1. Perform basic configurations on FW_A, including interface IP addresses, security zones to which the interfaces are added, interzone security policies, and static routes.
 - a. Set interface IP addresses.
 - i. Set the IP address of GE1/0/3.

```
<sysname> system-view
```

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```
[sysname] sysname FW_A
[FW_A] interface gigabitethernet 1/0/3
[FW_A-GigabitEthernet1/0/3] ip address 10.1.1.1 24
[FW_A-GigabitEthernet1/0/3] quit
```

2. Set the IP address of GE1/0/1.

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ip address 1.1.3.1 24
[FW_A-GigabitEthernet1/0/1] quit
```

3. Add interfaces to corresponding security zones.

Add GE1/0/3 to the Trust zone.

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface gigabitethernet 1/0/3
[FW_A-zone-trust] quit
```

4. Add GE1/0/1 to the Untrust zone.

```
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface gigabitethernet 1/0/1
[FW_A-zone-untrust] quit
```

5. Configure interzone security policies.

a. Configure the security policies between the Trust and Untrust zones.

```
[FW_A] security-policy
[FW_A-policy-security] rule name policy1
[FW_A-policy-security-rule-policy1] source-zone trust
[FW_A-policy-security-rule-policy1] destination-zone untrust
[FW_A-policy-security-rule-policy1] source-address 10.1.1.0 24
[FW_A-policy-security-rule-policy1] destination-address 10.1.2.0 24
[FW_A-policy-security-rule-policy1] action permit
[FW_A-policy-security-rule-policy1] quit
[FW_A-policy-security] rule name policy2
[FW_A-policy-security-rule-policy2] source-zone untrust
[FW_A-policy-security-rule-policy2] destination-zone trust
[FW_A-policy-security-rule-policy2] source-address 10.1.2.0 24
[FW_A-policy-security-rule-policy2] destination-address 10.1.1.0 24
[FW_A-policy-security-rule-policy2] action permit
[FW_A-policy-security-rule-policy2] quit
```

6. Configure the security policies between the Local and Untrust zones.



NOTE:

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The Local-Untrust interzone policy controls whether IKE negotiation packets can pass through the FW. This policy can use the source and destination addresses, protocol, or port as the matching condition. In this example, the source and destination addresses are used as the matching condition. To use the protocol or port as the matching condition, you need to enable ESP and port 500 for UDP (port 4500 also in NAT traversal scenarios).

Configure the security policies between the Local and Untrust zones to permit the interzone traffic for the negotiation between the tunnel endpoints.

```
[FW_A-policy-security] rule name policy3
[FW_A-policy-security-rule-policy3] source-zone local
[FW_A-policy-security-rule-policy3] destination-zone untrust
[FW_A-policy-security-rule-policy3] source-address 1.1.3.1 32
[FW_A-policy-security-rule-policy3] destination-address 1.1.5.1 32 [FW_A-
policy-security-rule-policy3] action permit
[FW_A-policy-security-rule-policy3] quit
[FW_A-policy-security] rule name policy4
[FW_A-policy-security-rule-policy4] source-zone untrust
[FW_A-policy-security-rule-policy4] destination-zone local
[FW_A-policy-security-rule-policy4] source-address 1.1.5.1 32
[FW_A-policy-security-rule-policy4] destination-address 1.1.3.1 32
[FW_A-policy-security-rule-policy4] action permit
[FW_A-policy-security-rule-policy4] quit
[FW_A-policy-security] quit
```

- a. Configure a static route to network B. Assume that the next hop of the route is 1.1.3.2.

```
[FW_A] ip route-static 1.1.5.0 255.255.255.0 1.1.3.2
```

7. Configure IPSec on FW_A.

- a. Configure IPSec tunnel interfaces.

```
[FW_A] interface Tunnel 1
[FW_A-Tunnel1] tunnel-protocol ipsec
[FW_A-Tunnel1] source 1.1.3.1
[FW_A-Tunnel1] destination 1.1.5.1
[FW_A-Tunnel1] ip address 172.16.2.1 24
[FW_A-Tunnel1] quit
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface tunnel 1
[FW_A-zone-untrust] quit
```

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- b. Configure a static route to direct traffic to the IPSec tunnel.

```
[FW_A] ip route-static 10.1.2.0 24 Tunnel1
```

- c. Configure an IPSec proposal. The default parameters may not be configured.

```
[FW_A] ipsec proposal tran1  
[FW_A-ipsec-proposal-tran1] esp authentication-algorithm sha2-256  
[FW_A-ipsec-proposal-tran1] esp encryption-algorithm aes-256  
[FW_A-ipsec-proposal-tran1] quit
```

- d. Configure an IKE proposal. The default parameters may not be configured.

```
[FW_A] ike proposal 10  
[FW_A-ike-proposal-10] authentication-method pre-share  
[FW_A-ike-proposal-10] prf hmac-sha2-256  
[FW_A-ike-proposal-10] encryption-algorithm aes-256  
[FW_A-ike-proposal-10] dh group14  
[FW_A-ike-proposal-10] integrity-algorithm hmac-sha2-256 [FW_A-ike-proposal-10]  
quit
```

- e. Configure an IKE peer.

```
[FW_A] ike peer b  
[FW_A-ike-peer-b] ike-proposal 10  
[FW_A-ike-peer-b] pre-shared-key Test!1234  
[FW_A-ike-peer-b] quit
```

- f. Configure an IPSec profile.

```
[FW_A] ipsec profile pro1  
[FW_A-ipsec-profile-pro1] proposal tran1  
[FW_A-ipsec-profile-pro1] ike-peer b  
[FW_A-ipsec-profile-pro1] quit
```

- g. Apply IPSec profile **pro1** to the IPSec tunnel interface.

```
[FW_A] interface tunnel 1  
[FW_A-Tunnel1] ipsec profile pro1  
[FW_A-Tunnel1] quit
```

8. Perform basic configurations on FW_B, including interface IP addresses, security zones to which the interfaces are added, interzone security policies, and static routes.

- a. Set interface IP addresses.

- i. Configure the IP address of GE1/0/3.

```
<sysname> system-view  
[sysname] sysname FW_B  
[FW_B] interface gigabitethernet 1/0/3  
[FW_B-GigabitEthernet1/0/3] ip address 10.1.2.1 24  
[FW_B-GigabitEthernet1/0/3] quit
```

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- ii. Configure the IP address of GE1/0/1.

```
[FW_B] interface gigabitethernet 1/0/1
[FW_B-GigabitEthernet1/0/1] ip address 1.1.5.1 24
[FW_B-GigabitEthernet1/0/1] quit
```

- b. Add interfaces to corresponding security zones.

- i. Add GE1/0/3 to the Trust zone.

```
[FW_B] firewall zone trust
[FW_B-zone-trust] add interface gigabitethernet 1/0/3
[FW_B-zone-trust] quit
```

- ii. Add GE1/0/1 to the Untrust zone.

```
[FW_B] firewall zone untrust
[FW_B-zone-untrust] add interface gigabitethernet 1/0/1
[FW_B-zone-untrust] quit
```

- c. Configure interzone security policies.

- i. Configure the security policies between the Trust and Untrust zones.

```
[FW_B] security-policy
[FW_B-policy-security] rule name policy1
[FW_B-policy-security-rule-policy1] source-zone trust
[FW_B-policy-security-rule-policy1] destination-zone untrust
[FW_B-policy-security-rule-policy1] source-address 10.1.2.0 24
[FW_B-policy-security-rule-policy1] destination-address 10.1.1.0 24
[FW_B-policy-security-rule-policy1] action permit
[FW_B-policy-security-rule-policy1] quit
[FW_B-policy-security] rule name policy2
[FW_B-policy-security-rule-policy2] source-zone untrust
[FW_B-policy-security-rule-policy2] destination-zone trust
[FW_B-policy-security-rule-policy2] source-address 10.1.1.0 24
[FW_B-policy-security-rule-policy2] destination-address 10.1.2.0 24
[FW_B-policy-security-rule-policy2] action permit
[FW_B-policy-security-rule-policy2] quit
```

- ii. Configure the security policies between the Local and Untrust zones.

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NOTE:

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Configure the security policies between the Local and Untrust zones to permit the interzone traffic for the negotiation between the tunnel endpoints.

```
[FW_B-policy-security] rule name policy3
[FW_B-policy-security-rule-policy3] source-zone local
[FW_B-policy-security-rule-policy3] destination-zone untrust
[FW_B-policy-security-rule-policy3] source-address 1.1.5.1 32
[FW_B-policy-security-rule-policy3] destination-address 1.1.3.1 32
[FW_B-policy-security-rule-policy3] action permit
[FW_B-policy-security-rule-policy3] quit
[FW_B-policy-security] rule name policy4
[FW_B-policy-security-rule-policy4] source-zone untrust
[FW_B-policy-security-rule-policy4] destination-zone local
[FW_B-policy-security-rule-policy4] source-address 1.1.3.1 32
[FW_B-policy-security-rule-policy4] destination-address 1.1.5.1 32
[FW_B-policy-security-rule-policy4] action permit
[FW_B-policy-security-rule-policy4] quit
[FW_B-policy-security] quit
```

- d. Configure a static route to network A. Assume that the next hop of the route is 1.1.5.2.

```
[FW_B] ip route-static 1.1.3.0 255.255.255.0 1.1.5.2
```

9. Configure IPSec on FW_B.

- a. Configure IPSec tunnel interfaces.

```
[FW_B] interface Tunnel 1
[FW_B-Tunnel1] tunnel-protocol ipsec
[FW_B-Tunnel1] source 1.1.5.1
[FW_B-Tunnel1] destination 1.1.3.1
[FW_B-Tunnel1] ip address 172.16.2.2 24
[FW_B-Tunnel1] quit
[FW_B] firewall zone untrust
```

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```
[FW_B-zone-untrust] add interface tunnel 1  
[FW_B-zone-untrust] quit
```

- b. Configure a static route to direct traffic to the IPSec tunnel.

```
[FW_B] ip route-static 10.1.1.0 24 Tunnel1
```

- c. Configure an IPSec proposal.

```
[FW_B] ipsec proposal tran1  
[FW_B-ipsec-proposal-tran1] esp authentication-algorithm sha2-256  
[FW_B-ipsec-proposal-tran1] esp encryption-algorithm aes-256  
[FW_B-ipsec-proposal-tran1] quit
```

- d. Configure an IKE proposal.

```
[FW_B] ike proposal 10  
[FW_B-ike-proposal-10] authentication-method pre-share  
[FW_B-ike-proposal-10] prf hmac-sha2-256  
[FW_B-ike-proposal-10] encryption-algorithm aes-256  
[FW_B-ike-proposal-10] dh group14  
[FW_B-ike-proposal-10] integrity-algorithm hmac-sha2-256  
[FW_B-ike-proposal-10] quit
```

- e. Configure an IKE peer.

```
[FW_B] ike peer a  
[FW_B-ike-peer-a] ike-proposal 10  
[FW_B-ike-peer-a] pre-shared-key Test!1234  
[FW_B-ike-peer-a] quit
```

- f. Configure an IPSec profile.

```
[FW_B] ipsec profile pro1  
[FW_B-ipsec-profile-pro1] proposal tran1  
[FW_B-ipsec-profile-pro1] ike-peer a  
[FW_B-ipsec-profile-pro1] quit
```

- g. Apply IPSec profile **pro1** to a tunnel interface.

```
[FW_B] interface tunnel 1  
[FW_B-Tunnel1] ipsec profile pro1 [FW_B-Tunnel1] quit
```

Verification

Run the **display ike sa** and **display ipsec sa** commands on both FW_A and FW_B to check SA establishment.

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Take FW_B for example. If the following information is displayed, the IKE SA and IPSec SA are successfully established.

```
<FW_B> display ike sa
```

Ike sa information :

Conn-ID	Peer	VPN	Flag(s)	Phase
16782310	1.1.3.1		RD ST A	v2:2
16782309	1.1.3.1		RD ST A	v2:1

Number of SA entries : 2

Number of SA entries of all cpu : 2

Flag Description:

RD--READY ST--STAYALIVE RL--REPLACED FD--FADING TO--TIMEOUT
HRT--HEARTBEAT LKG--LAST KNOWN GOOD SEQ NO. BCK--BACKED UP
M--ACTIVE S--STANDBY A--ALONE NEG--NEGOTIATING

```
<FW_B> display ipsec sa
```

ipsec sa information:

=====

Interface: Tunnel1

=====

IPSec profile name: "pro1"

Mode : PROF-ISAKMP

Connection ID : 16782310

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```
Encapsulation mode: Tunnel
Tunnel local      : 1.1.5.1
Tunnel remote     : 1.1.3.1

[Outbound ESP SAs]
SPI: 3174037361 (0xbd2ff771)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA remaining key duration (kilobytes/sec): 20971520/3551
Max sent sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA encrypted packets (number/bytes): 0/0

[Inbound ESP SAs]
SPI: 867840252 (0x33ba30fc)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA remaining key duration (kilobytes/sec): 20971520/3551
Max received sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA decrypted packets (number/bytes): 0/0
Anti-replay : Disable
```

Task:

Attach Screenshots of all the topology and configuration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. What is the main difference between a route-based IPSec VPN and a policy-based IPSec VPN?

Answer:

Route-based VPN uses a virtual tunnel interface and relies on static routes to direct traffic into the tunnel, while policy-based VPN uses access control lists (ACLs) or security policies to match and encrypt specific traffic between source and destination networks.

2. Why is a pre-shared key used in IPSec VPN authentication, and what are its security implications?

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Answer:

A pre-shared key (PSK) is used for simple, symmetric authentication between VPN peers without requiring a PKI. **Security implications:** It must be kept secret; weak or hardcoded keys are vulnerable to brute-force attacks, and key distribution is manual, making it less scalable and secure than certificate-based authentication.

3. Which protocols are used in the IPSec framework to ensure data confidentiality and integrity?

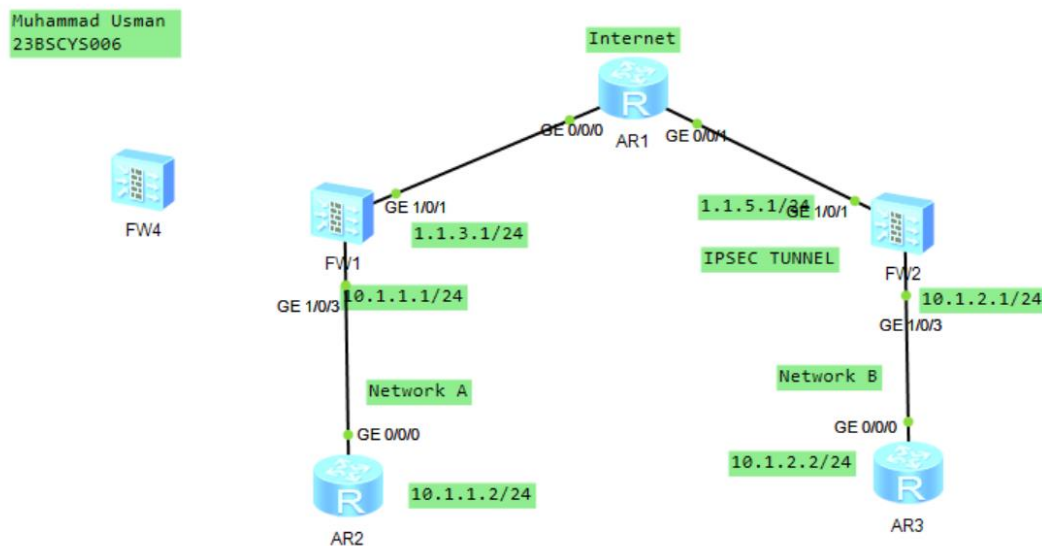
Answer:

- **Confidentiality (encryption):** ESP (Encapsulating Security Protocol) using algorithms like AES-256.
- **Integrity (hashing):** ESP and/or AH (Authentication Header) using algorithms like SHA2-256.
- **Key management & secure association:** IKE (Internet Key Exchange) using UDP port 500/4500.

4. What command or method can be used to test connectivity between hosts on Network A and Network B once the VPN tunnel is active?

Answer:

Use the **ping** command with source and destination IP addresses from each network, e.g.:
ping -a 10.1.1.1 10.1.2.1 (from FW_A to simulate host on Network A pinging Network B).
Alternatively, use extended ping or actual host-to-host ICMP echo requests.



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```
[23BSCYS006-FW-A] dis ike sa
2026-04-11 15:03:03.680

IKE SA information :
Conn-ID      Peer          Phase RemoteType RemoteID      VPN      Flag(s)
-----
6            1.1.5.1:500              v2:2  IP        1.1.5.1      RD|A
5            1.1.5.1:500              v2:1  IP        1.1.5.1      RD|A

Number of IKE SA : 2

Flag Description:
RD--READY  ST--STAYALIVE  RL--REPLACED  FD--FADING  TO--TIMEOUT
HRT--HEARTBEAT  LKG--LAST KNOWN GOOD SEQ NO.  BCK--BACKED UP
M--ACTIVE  S--STANDBY  A--ALONE  NEG--NEGOTIATING

[23BSCYS006-FW-A]dis ipsec sa
2026-04-11 15:03:08.120

[23BSCYS006-FW-A]dis ipsec sa
2026-04-11 15:03:08.120

ipsec sa information:
=====
Interface: Tunnell
=====

IPSec profile name: "prol"
Mode                : PROF-ISAKMP
=====

Connection ID       : 6
Encapsulation mode: Tunnel
Holding time        : 0d 0h 8m 21s
Tunnel local        : 1.1.3.1:500
Tunnel remote       : 1.1.5.1:500
Flow source         : 0.0.0.0/0.0.0.0 0/0-65535
Flow destination    : 0.0.0.0/0.0.0.0 0/0-65535

[Outbound ESP SAs]
SPI: 193404892 (0xb871fdc)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA remaining key duration (kilobytes/sec): 10485760/3099
Max sent sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA encrypted packets (number/bytes): 0/0

[Inbound ESP SAs]
SPI: 191999585 (0xb71ae61)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA remaining key duration (kilobytes/sec): 10485760/3099
Max received sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA decrypted packets (number/bytes): 0/0
Anti-replay : Enable
Anti-replay window size: 1024
[23BSCYS006-FW-A]
```

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```
[23BSCYS006-FW-B]dis ike sa
2026-04-11 15:04:50.710

IKE SA information :
Conn-ID   Peer          Phase RemoteType RemoteID      VPN      Flag(
s)
-----
6         1.1.3.1:500          v2:2  IP        1.1.3.1      RD|ST
|A
5         1.1.3.1:500          v2:1  IP        1.1.3.1      RD|ST
|A

Number of IKE SA : 2
-----

Flag Description:
RD--READY   ST--STAYALIVE  RL--REPLACED  FD--FADING   TO--TIMEOUT
HRT--HEARTBEAT  LKG--LAST KNOWN GOOD SEQ NO.  BCK--BACKED UP
M--ACTIVE    S--STANDBY    A--ALONE     NEG--NEGOTIATING

[23BSCYS006-FW-B]
```

FW1
AR3
FW2

```
[23BSCYS006-FW-B]dis ipsec sa
2026-04-11 15:05:13.130

ipsec sa information:
=====
Interface: Tunnel1
=====

-----
IPSec profile name: "pro1"
Mode                : PROF-ISAKMP
-----

Connection ID       : 6
Encapsulation mode: Tunnel
Holding time        : 0d 0h 10m 7s
Tunnel local        : 1.1.5.1:500
Tunnel remote       : 1.1.3.1:500
Flow source         : 0.0.0.0/0.0.0.0 0/0-65535
Flow destination    : 0.0.0.0/0.0.0.0 0/0-65535

[Outbound ESP SAs]
SPI: 191999585 (0xb71ae61)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA remaining key duration (kilobytes/sec): 10485760/2993
Max sent sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA encrypted packets (number/bytes): 0/0

[Inbound ESP SAs]
SPI: 193404892 (0xb871fdc)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA remaining key duration (kilobytes/sec): 10485760/2993
Max received sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA decrypted packets (number/bytes): 0/0
Anti-replay : Enable
Anti-replay window size: 1024

[23BSCYS006-FW-B]
```